DiffFNO: Fourier-Structured Diffusion Control for IoT-Enabled Building Energy Management

Wei Zou, Zongwei Han, Bo Cheng, Yu Sun, Jingbo Liu, Zongyi Liu, and Haibo Zhou
School of Electronic Science and Engineering, Nanjing University, Nanjing, China, 210023
Emails: {weizou,502025230079,bocheng,yusun,zongyiliu}@smail.nju.edu.cn, {jingboliu,haibozhou}@nju.edu.cn

***Abstract*—Internet of Things (IoT) sensor networks enable real-time thermal monitoring and control command delivery, making data-driven heating, ventilation, and air conditioning (HVAC) optimization promising for building energy management. However, IoT-enabled building energy management remains challenging because real-time sensor streams must be converted into coordinated control actions that balance energy saving and comfort compliance under coupled thermal dynamics. Legacy reinforcement learning policies often rely on restrictive unimodal action models, multilayer perceptron (MLP)-based policies do not explicitly model zone-wise coordination, and standard learned policies may produce oscillatory actions that reduce control smoothness and weaken physical consistency. To this end, we present DiffFNO, a generative controller for IoT-enabled building energy management that integrates a conditional diffusion policy, an action-dimensional Fourier neural operator (FNO) denoiser, and a residual correction pathway. Simulation results on the Physics-Principled Building Environment for Control and Reinforcement Learning (BEAR) OfficeSmall-Hot_Dry benchmark demonstrate its effectiveness in improving the energy–comfort trade-off and producing smoother, more physically consistent control signals. Ablation studies further support the contribution of the residual pathway, suggesting the potential of DiffFNO for IoT-enabled building energy management.**

***Index Terms*—IoT-enabled building energy management, HVAC optimization, diffusion control, Fourier neural operator, reinforcement learning.**

I. Introduction

Buildings account for a substantial portion of global energy consumption, and heating, ventilation, and air conditioning (HVAC) systems are among the largest controllable contributors to operational energy use [1], [2]. In Internet of Things (IoT)-enabled buildings, distributed sensors continuously monitor zone temperatures, outdoor conditions, solar irradiance, and occupancy-related loads. These measurements are aggregated through gateway nodes and fed into the controller at each decision step. Recent IoT and industrial IoT studies have also emphasized that sensing-data aggregation, information freshness, and edge-side processing are central to reliable closed-loop services [3], [4]. The resulting control actions are then applied to zone-level HVAC actuators through the same IoT infrastructure, forming the sensing–actuation loop. This closed-loop digital infrastructure makes data-driven HVAC control increasingly relevant for large commercial buildings.

In occupied buildings, persistent thermal-comfort violations are not merely a secondary penalty. They directly affect whether an energy-saving controller is acceptable to occupants and facility operators. Repeated comfort-band violations can lead to occupant discomfort, manual override of automated control actions, and loss of deployment trust. Energy efficiency is therefore best viewed as a constrained operating objective: a controller that saves energy by tolerating chronic overheating or overcooling is not practically deployable. This perspective motivates treating comfort violations as a primary evaluation criterion alongside energy consumption.

Recent reinforcement learning (RL) approaches provide a model-free alternative to rule-based control and model predictive control (MPC), but standard policies remain mismatched to multi-zone HVAC control in three ways [5]–[8]. First, many continuous-control methods use deterministic mappings or unimodal Gaussian action distributions, which are too restrictive when multiple coordinated control patterns are feasible under the same thermal state. Second, common multilayer perceptron (MLP)-based policies do not explicitly encode cross-zone coupling and therefore struggle to capture the structured geometry of the joint action vector. Third, standard learned policies may produce oscillatory actions that reduce control smoothness and weaken physical consistency in coupled building systems.

Diffusion policies offer a more expressive action model for continuous control [9]–[11], while Fourier neural operators (FNOs) provide a useful spectral prior when low-frequency coordination is important [12], [13]. Motivated by these observations, we formulate the multi-zone HVAC policy as a conditional reverse-diffusion process and replace the standard MLP denoiser with an FNO denoiser that acts along the action dimension. We further retain a residual pathway so the denoiser can preserve coarse spectral coordination while still making local corrective adjustments when comfort boundaries are tight. Intuitively, the action-dimension axis can be viewed as a compact control field over thermal zones: neighboring or strongly coupled zones should usually receive coordinated commands, while abrupt high-frequency variations across action coordinates often indicate physically inconsistent control. This makes an FNO-based spectral prior well aligned with multi-zone HVAC actions. The policy class should therefore be able to represent multiple feasible joint actions, preserve cross-zone consistency, and remain stable under repeated sensor-driven updates.

Unlike conventional building RL policies that predict actions coordinate-wise, DiffFNO treats the full multi-zone command as a generated object. It also differs from generic diffusion control by embedding building-specific action-space

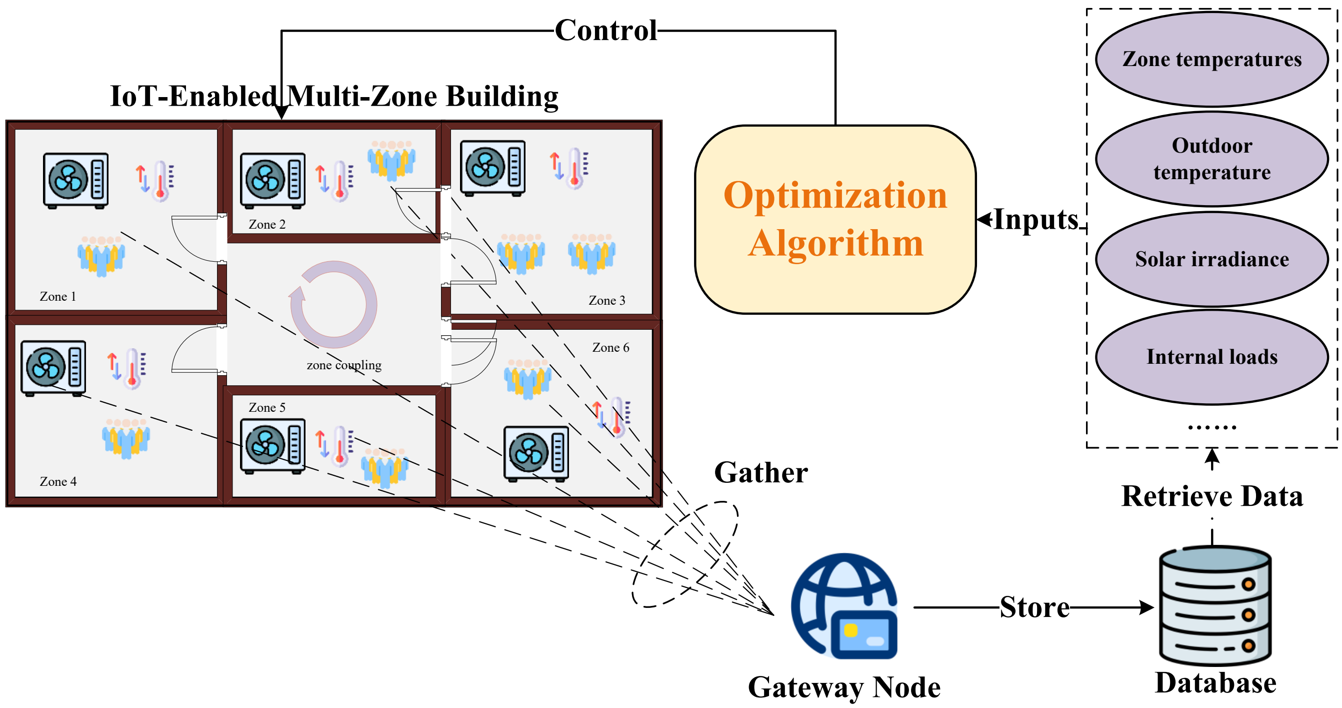

Fig. 1: IoT-enabled sensing–control architecture for multi-zone building HVAC control. Zone-level measurements are gathered by the gateway node, aggregated into the state vector $\mathbf{s}_t$, and fed to the controller, whose joint action is applied back to the building.

structure through the FNO denoiser and residual correction pathway.

The main contributions of this work are:

- We propose *DiffFNO*, a Fourier-structured generative controller for IoT-enabled building energy management that models the full joint zone action through conditional diffusion rather than point-estimate regression.
- We introduce an FNO denoiser operating along the action-dimension axis—treating the joint zone command as a control field rather than a temporal sequence—together with a gated residual pathway that preserves local corrective flexibility near comfort boundaries.
- We show on the Physics-Principled Building Environment for Control and Reinforcement Learning (BEAR) benchmark that DiffFNO improves the energy–comfort profile over Diffusion-MLP, and that removing the residual pathway yields lower energy and smoother actions only at the cost of increased comfort violations.

The remainder of this paper is organized as follows. Section II introduces the multi-zone HVAC control model. Section III presents the proposed DiffFNO framework. Section IV reports the experimental setup and results. Finally, Section V concludes the paper and discusses future extensions.

II. System Model

We consider an IoT-enabled multi-zone HVAC control task in which a supervisory controller maps building-wide sensor measurements to coordinated zone-level commands, with the objective of reducing energy use while avoiding thermal-comfort violations. Figure 1 illustrates the corresponding sensing–control loop: zone-level measurements are gathered through the gateway node, aggregated into the state vector $\mathbf{s}_t$, and fed to the controller. The resulting joint action is then applied back to the zone-level HVAC actuators. Such sensing–actuation loops are consistent with recent communication-control co-design studies in industrial IoT systems, where transmission and control performance are jointly considered [14].

We formulate multi-zone HVAC control as a discounted Markov decision process (MDP) $\mathcal{M} = (\mathcal{S}, \mathcal{A}, \mathcal{P}, r, \gamma)$, where the transition kernel is induced by the physics-based BEAR simulator [15]. The controller interacts with a six-zone office building under coupled thermal dynamics, weather disturbances, solar gains, and internal loads. The state vector $\mathbf{s}_t$ is constructed from aggregated IoT measurements collected across all thermal zones, including zone temperatures, outdoor temperature, solar irradiance, ground temperature, and internal loads. In Fig. 1, the retrieved-data block highlights the main components of $\mathbf{s}_t$, while the ellipsis denotes additional exogenous features such as the ground-temperature term. The action is a continuous joint HVAC command for all N zones:

$$\mathbf{a}_t = [a_{t,1}, a_{t,2}, \ldots, a_{t,N}] \in [-1, 1]^N, \tag{1}$$

so the policy must generate a coordinated multi-zone control vector rather than independent per-zone decisions. Because the six zones share disturbances and thermal interactions, a command that appears locally beneficial for one zone may still be globally inefficient once neighboring temperatures, comfort violations, and total HVAC energy are considered jointly.

The reward combines the simulator reward with an explicit thermal-comfort violation penalty:

$$d_{t,i} = \left|T_{t,i}^{\text{zone}} - T^{\text{ref}}\right|, \qquad V_t = \sum_{i=1}^{N} \mathbb{I}(d_{t,i} > \delta), \tag{2}$$

$$r_t = \rho \left(r_t^{\text{env}} - \lambda V_t\right), \tag{3}$$

where T^{ref} is the target temperature, δ is the comfort tolerance, V_t counts zone-wise temperature violations with respect to the comfort band defined by the American Society of Heating, Refrigerating and Air-Conditioning Engineers (ASHRAE) Standard 55 [16], λ is the violation penalty, and ρ is a reward-scaling coefficient. The control objective is

$$J(\pi) = \mathbb{E}_{\pi}\left[\sum_{t=0}^{T-1} \gamma^t r_t\right]. \tag{4}$$

This objective matches the control target of reducing HVAC energy while avoiding persistent comfort-band violations across the monitored thermal zones.

III. DiffFNO Method

A. Conditional Diffusion Policy

Instead of regressing a single action from $\mathbf{s}_t$, we generate actions through a conditional denoising process [17]. Let $\mathbf{x}_0$ denote the clean action and $\mathbf{x}_k$ its noisy version at reverse step k. Under the variance-preserving parameterization,

$$\mathbf{x}_k = \sqrt{\bar{\alpha}_k}\mathbf{x}_0 + \sqrt{1-\bar{\alpha}_k}\boldsymbol{\epsilon}, \qquad \boldsymbol{\epsilon} \sim \mathcal{N}(\mathbf{0}, \mathbf{I}), \tag{5}$$

and the denoiser directly predicts the clean action

$$\hat{\mathbf{x}}_0 = f_\theta(\mathbf{x}_k, k, \mathbf{s}_t). \tag{6}$$

Under the standard denoising diffusion probabilistic model (DDPM) parameterization, the reverse posterior mean can be written as

$$\boldsymbol{\mu}_k(\hat{\mathbf{x}}_0, \mathbf{x}_k) = c_1(k)\hat{\mathbf{x}}_0 + c_2(k)\mathbf{x}_k, \tag{7}$$

where $c_1(k)$ and $c_2(k)$ are determined by the noise schedule. Starting from Gaussian noise, the policy iteratively reconstructs a feasible multi-zone action through reverse DDPM sampling. This formulation is attractive for HVAC control because it can represent richer action distributions than deterministic or unimodal Gaussian policies while exposing the structure of the joint action to the denoiser.

B. Fourier-Structured Action Denoiser

The denoiser is the key component that injects action-space structure into the diffusion controller. In multi-zone HVAC control, the joint action should not be viewed as an unordered list of independent coordinates, because zone-level commands are coupled through shared thermal dynamics and comfort constraints. A standard MLP denoiser has limited ability to express this structured coordination explicitly. We therefore

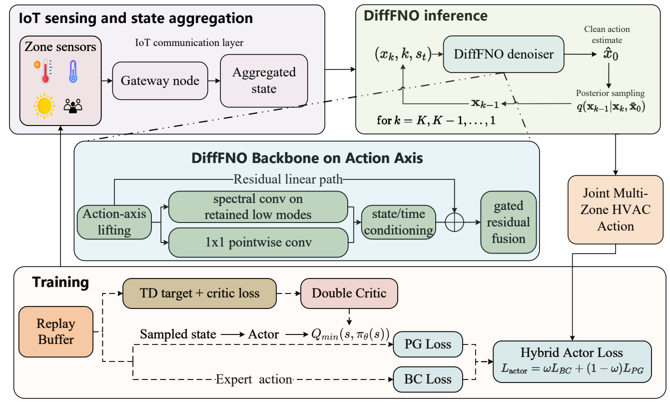

Fig. 2: DiffFNO framework. The policy models the joint action through reverse diffusion, the DiffFNO denoiser imposes spectral structure on the action-dimension axis, and the residual pathway preserves local corrective flexibility. A double critic is used during training, while deployment uses reverse diffusion sampling.

replace the MLP denoiser with an FNO denoiser that operates directly along the action-dimension axis, so that the joint action is processed as a compact control field over zones. Given noisy action $\mathbf{x}_k$, diffusion step k, and state $\mathbf{s}_t$, the latent feature map is updated by

$$\mathbf{Y}^{(\ell+1)} = \sigma\left(\mathcal{K}_{\text{spec}}^{(\ell)}(\mathbf{Y}^{(\ell)}) + \mathcal{K}_{\text{pw}}^{(\ell)}(\mathbf{Y}^{(\ell)}) + \mathbf{h}_c\right), \tag{8}$$

where $\mathcal{K}_{\text{spec}}^{(\ell)}$ is a truncated one-dimensional spectral convolution, $\mathcal{K}_{\text{pw}}^{(\ell)}$ is a pointwise convolution, and $\mathbf{h}_c$ is the joint embedding of state and diffusion-step information. The final clean-action estimate is

$$f_\theta(\mathbf{x}_k, k, \mathbf{s}_t) = \hat{\mathbf{y}}_k + \text{sigmoid}(g)\mathbf{W}_r\mathbf{x}_k. \tag{9}$$

Here $\hat{\mathbf{y}}_k$ is the spectrally structured output of the FNO trunk and the gated residual term re-injects local coordinate information from the noisy action. This design encourages globally coordinated low-frequency action patterns while retaining enough local flexibility for zone-specific comfort regulation. In the deployment setting studied here, this residual pathway is an essential part of the main method because it improves comfort compliance relative to the no-residual alternative.

C. Hybrid Training Framework

Although actions are generated through reverse diffusion sampling at deployment, the actor is still trained within an off-policy actor–critic framework. We use a conservative double critic

$$Q_\phi^{\min}(\mathbf{s}, \mathbf{a}) = \min\{Q_{\phi_1}(\mathbf{s}, \mathbf{a}), Q_{\phi_2}(\mathbf{s}, \mathbf{a})\}, \tag{10}$$

for policy evaluation during training. The actor combines diffusion-consistent behavior cloning (BC) from MPC demonstrations with critic-driven policy gradient (PG) improvement:

$$\mathcal{L}_{\text{BC}} = \mathbb{E}\left[\|f_\theta(\mathbf{x}_k^{\text{exp}}, k, \mathbf{s}_t) - \mathbf{a}_t^{\text{exp}}\|_2^2\right], \tag{11}$$

$$\mathcal{L}_{\text{PG}} = -\mathbb{E}_{\mathbf{s}_t}\left[Q_\phi^{\min}(\mathbf{s}_t, \pi_\theta(\mathbf{s}_t))\right], \tag{12}$$

$$\mathcal{L}_{\text{actor}} = \omega \mathcal{L}_{\text{BC}} + (1 - \omega)\mathcal{L}_{\text{PG}}, \tag{13}$$

where $\mathcal{L}_{\text{BC}}$ reconstructs expert actions from noisy inputs and $\mathcal{L}_{\text{PG}}$ maximizes the conservative critic value [18]. A high initial ω stabilizes early learning, and the weight is then decayed so that the policy can move beyond imitation while preserving physically meaningful action priors. This hybrid design allows the actor to inherit low-jitter control behavior from MPC before refining the policy with long-horizon value information. At deployment time, the controller executes the learned DiffFNO policy through reverse diffusion sampling.

Under the centralized IoT interface considered here, building-wide measurements are aggregated into a common state vector, and the controller must map that shared state to a coordinated joint action. This setting directly motivates a state-conditional generative policy with explicit action-space structure.

This centralized joint-action view is complementary to decentralized and multi-agent HVAC formulations. Multi-agent methods can reduce per-agent action dimensionality and support local autonomy, but they often require additional coordination mechanisms, communication assumptions, or reward-allocation designs to recover system-level energy–comfort objectives [19]–[21]. In contrast, the setting considered here assumes that an IoT gateway already aggregates building-wide sensor streams for supervisory control. Under this interface, it is natural to let one policy generate the full zone-wise HVAC command while embedding the desired coordination directly in the action representation. The contribution is therefore architectural rather than organizational: DiffFNO does not change the MDP or add handcrafted zone rules, but changes how the policy represents and denoises the coupled action vector.

IV. Simulation Experiments and Discussions

A. Setup

We evaluate all methods on the BEAR OfficeSmall-Hot_Dry benchmark [15], which models a six-zone office building under Tucson hot-dry weather. The controller receives synchronized sensor-style state streams at each control step and returns zone-wise HVAC commands through the same interface. The control interval is one hour and each episode lasts 168 steps. Unless otherwise stated, all methods share the same target temperature, comfort band, simulator disturbances, and reward setting, which helps isolate controller quality rather than task-definition changes. The main comparison focuses on learning-based controllers, including DiffFNO, DiffFNO w/o Residual, Diffusion-MLP, soft actor-critic (SAC), and SAC+MPC. The two DiffFNO variants share the same diffusion training and deployment pipeline and differ only in whether the residual pathway is enabled, while Diffusion-MLP replaces the spectral denoiser with a standard MLP denoiser. MPC with planning horizon 3 is used to generate expert demonstrations and serves as a physical reference in the spectral analysis. Learning-based methods report three-seed statistics to reflect training variability. We report four complementary metrics: episode

TABLE I: Shared hyperparameters for the learning-based methods.

Setting	Value	Setting	Value
Training steps	1.0×10^6	Batch size / replay buffer	256 / 10^6
Discount / n-step return	0.99 / 3	Update-per-step	0.5
Actor / critic learning rate	10^{-4} / 2×10^{-5}	Diffusion steps K	6
FNO width / retained modes	48 / 4	Spectral layers / activation	1 / Mish
Main residual pathway	enabled	BC weight schedule ω	$0.8 \rightarrow 0.1$

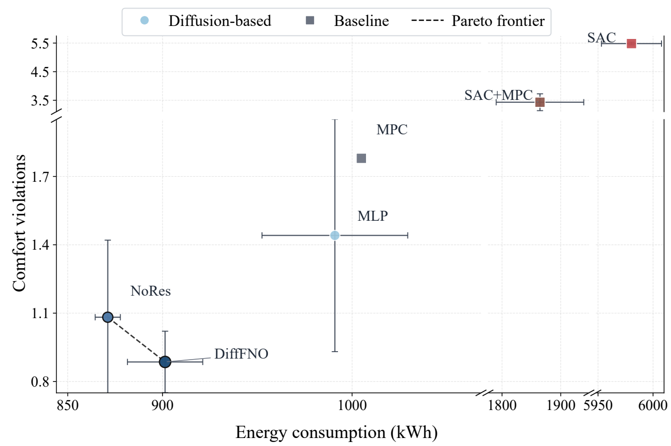

Fig. 3: Energy–comfort Pareto comparison for the evaluated controllers. Learning-based points show three-seed means and error bars show one standard deviation, while MPC is included as a physical reference controller.

HVAC energy, comfort violations, final test reward, and mean absolute action change. Energy and comfort violations jointly reflect the primary deployment objective, while final reward and mean absolute action change capture the optimized RL objective and control smoothness.

Inference cost. DiffFNO uses only $K = 6$ reverse diffusion steps at deployment. On a single NVIDIA RTX 3070 graphics processing unit (GPU) with batch size 1, this corresponds to approximately 5.2 ms per denoising step, or about 31 ms per control decision for the six-step chain. The resulting online cost remains far below the one-hour control interval used in BEAR.

B. Overall Performance

Table II first highlights the main comparison in this study: DiffFNO versus the MLP-denoiser diffusion baseline. Under the same diffusion-control setting, replacing the MLP denoiser with the proposed DiffFNO architecture improves all four reported criteria. Relative to Diffusion-MLP, DiffFNO reduces episode energy from 990.9 to 901.3 kWh, lowers comfort violations from 1.441 to 0.886, improves the final test reward from -1.106 to -0.738, and reduces mean $|\Delta a|$ from 0.0596 to 0.0402. These gains indicate that the structured FNO denoiser and residual pathway make diffusion-based HVAC control more effective than a generic MLP denoiser.

The Pareto view in Fig. 3 reinforces the same conclusion. DiffFNO dominates Diffusion-MLP in both energy and comfort violations, showing that expressive diffusion alone is not sufficient without a denoiser that respects the structured

TABLE II: Overall comparison of learning-based controllers on the BEAR OfficeSmall-Hot_Dry task. Results report the three-seed mean $\pm$ standard deviation. Lower mean $|\Delta a|$ indicates smoother control.

Method	Energy (kWh) ↓	Violations ↓	Test Reward ↑	Mean $\lvert\Delta a\rvert$ ↓
DiffFNO	901.3 ± 19.9	**0.886 ± 0.135**	**-0.738 ± 0.086**	0.0402 ± 0.0090
DiffFNO w/o Residual	**871.1 ± 6.6**	1.082 ± 0.338	-0.844 ± 0.207	**0.0357 ± 0.0109**
Diffusion-MLP	990.9 ± 38.4	1.441 ± 0.510	-1.106 ± 0.321	0.0596 ± 0.0162
SAC	5980.4 ± 27.3	5.483 ± 0.046	-5.055 ± 0.011	0.3430 ± 0.0412
SAC+MPC	1864.6 ± 74.6	3.431 ± 0.294	-2.600 ± 0.222	0.1729 ± 0.0117

geometry of multi-zone actions. SAC-based baselines remain far from the relevant operating regime for this benchmark. MPC is retained in the Pareto view only as a physical reference controller, while Table II focuses on learning-based controllers. The no-residual variant is retained in this figure for component analysis, where its role is to probe the residual pathway rather than to define the primary baseline comparison.

Deployment implications. The results should be read as evidence for a policy-design principle rather than as a claim that one scalar metric is universally dominant. DiffFNO does not minimize raw energy among all variants. Instead, it represents a more comfort-compliant operating point while still improving energy, reward, and smoothness over the MLP-denoiser baseline. This distinction matters in occupied buildings, where persistent comfort violations can invalidate an apparently energy-efficient policy. The current study is also bounded by the OfficeSmall-Hot_Dry configuration, hourly control interval, and available MPC demonstrations. Extending the same principle to larger buildings may require topology-aware action ordering or graph-structured variants, but the present results show that even a compact spectral prior over the joint action vector can improve the behavior of diffusion-based HVAC control.

C. Action Quality and Ablation Study

The spectral prior imposed by DiffFNO is reflected in the physical HVAC signal. As shown in Fig. 4, DiffFNO concentrates substantially more spectral mass in the low-frequency region than Diffusion-MLP and moves closer to the smooth operating pattern of MPC. Using the 2 cycles/day cutoff in the Welch spectrum, DiffFNO assigns 82.6% of HVAC power spectral mass to the low-frequency region, compared with 95.2% for physical MPC and 73.7% for Diffusion-MLP. The result is consistent with the lower mean $|\Delta a|$ of DiffFNO relative to Diffusion-MLP in Table II: the proposed controller improves the overall control objective while generating a more physically consistent control signal than the MLP-denoiser baseline.

The spectral analysis is complemented by the ablation summary in Fig. 5, which clarifies the role of each module in the proposed controller. The no-residual variant should be interpreted as a component ablation: removing the residual pathway yields lower energy and smoother action increments, but it also increases comfort violations from 0.886 to 1.082 and reduces the final reward from -0.738 to -0.844. This pattern supports the interpretation that the residual pathway restores useful local flexibility near comfort boundaries, where

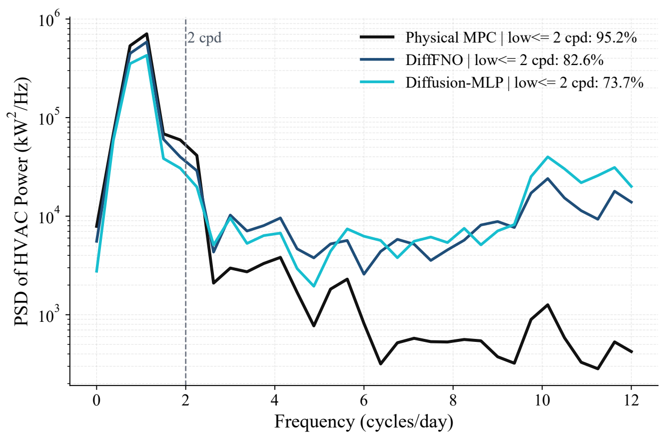

Fig. 4: HVAC power spectral density comparison for physical MPC, DiffFNO, and Diffusion-MLP. The learned-controller curves are averaged over three seeds. The dashed line marks the 2 cycles/day cutoff.

overly smooth coordinated actions may underreact to zone-specific thermal deviations. In Fig. 5, the “Convergence” column is computed from the normalized terminal evaluation reward across ablation variants, so it should be read as a compact terminal-performance indicator rather than as a claim about the full dynamic convergence trajectory. From a control perspective, building HVAC decisions are repeatedly executed over long horizons, and a policy that reduces energy but sacrifices comfort compliance may still be undesirable under coupled thermal dynamics. The smoother low-frequency behavior of DiffFNO therefore matters not only as a diagnostic statistic but also as evidence that the learned controller is better aligned with physically consistent building operation.

Taken together, the experiments suggest a clear division of labor among the model components. Diffusion provides expressive joint-action generation, DiffFNO injects a structural prior for coordinated multi-zone control, and the residual pathway improves comfort-oriented local correction. This decomposition matches the empirical pattern in Table II and Fig. 5, where no single simplified variant preserves all three advantages at once. The proposed design does not rely on handcrafted rules tied to a specific room or actuator. Instead, it relies on a generic centralized interface: a vector of sensor-derived building states, a vector of zone-wise control commands, and a reward that trades energy against comfort compliance. As long as that interface is available, the same modeling principle can be applied to other multi-zone buildings, climates, or supervisory control layers, although the final quantitative gains should still be validated case by case.

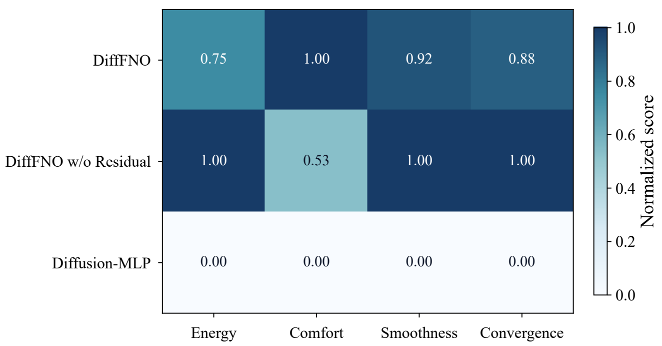

Fig. 5: Normalized ablation summary across energy, comfort, smoothness, and convergence for the evaluated diffusion variants. Each column is min–max normalized over DiffFNO, DiffFNO w/o Residual, and Diffusion-MLP.

V. Conclusion

This paper has presented DiffFNO for IoT-enabled building energy management and has evaluated it on the BEAR OfficeSmall-Hot_Dry benchmark. The simulation results have shown that, compared with the MLP-denoiser diffusion baseline, DiffFNO achieves a more favorable energy–comfort trade-off, reduces comfort violations, improves the final test reward, and generates smoother control trajectories. The ablation results have further shown that the residual pathway is important for comfort-oriented local correction: removing it lowers energy and action variation, but increases comfort violations and degrades reward. These findings suggest that structured action modeling is beneficial for IoT-enabled building control, especially when coordinated actuation and local comfort constraints must be handled simultaneously. Future directions include exploring value-aware sampling strategies, broader validation across additional building types and climate profiles, and extensions toward safety-aware control under deployment uncertainty.

References

[1] L. Pérez-Lombard, J. Ortiz, and C. Pout, "A review on buildings energy consumption information," *Energy and Buildings*, vol. 40, no. 3, pp. 394–398, 2008.

[2] X. Cao, X. Dai, and J. Liu, "Building energy-consumption status worldwide and the state-of-the-art technologies for zero-energy buildings during the past decade," *Energy and Buildings*, vol. 128, pp. 198–213, 2016.

[3] J. Wang, L. Bai, J. Chen, and J. Wang, "Age of information oriented data aggregation for industrial internet of things," in *2024 IEEE/CIC International Conference on Communications in China (ICCC)*. IEEE, 2024, pp. 621–626.

[4] H. Hou, S. Bi, X. Li, and X. Lin, "Joint semantic information gathering and broadcasting for multi-task edge processing," in *2024 IEEE/CIC International Conference on Communications in China (ICCC)*. IEEE, 2024, pp. 60–65.

[5] J. Drgoňa, J. Arroyo, I. Cupeiro Figueroa, D. Blum, K. Arendt, D. Kim, E. P. Ollé, J. Oravec, M. Wetter, D. L. Vrabie, and L. Helsen, "All you need to know about model predictive control for buildings," *Annual Reviews in Control*, vol. 50, pp. 190–232, 2020.

[6] T. Wei, Y. Wang, and Q. Zhu, "Deep reinforcement learning for building HVAC control," in *Proceedings of the 54th Annual Design Automation Conference 2017*. ACM, 2017, pp. 1–6.

[7] M. Biemann, F. Scheller, X. Liu, and L. Huang, "Experimental evaluation of model-free reinforcement learning algorithms for continuous HVAC control," *Applied Energy*, vol. 298, p. 117164, 2021.

[8] K. Al Sayed, A. Boodi, R. Sadeghian Broujeny, and K. Beddiar, "Reinforcement learning for HVAC control in intelligent buildings: A technical and conceptual review," *Journal of Building Engineering*, vol. 95, p. 110085, 2024.

[9] C. Chi, Z. Xu, S. Feng, E. Cousineau, Y. Du, B. Burchfiel, R. Tedrake, and S. Song, "Diffusion policy: Visuomotor policy learning via action diffusion," *The International Journal of Robotics Research*, vol. 44, no. 10–11, pp. 1684–1704, 2025.

[10] M. Janner, Y. Du, J. Tenenbaum, and S. Levine, "Planning with diffusion for flexible behavior synthesis," in *Proceedings of the 39th International Conference on Machine Learning*, vol. 162. PMLR, 2022, pp. 9902–9915.

[11] Z. Wang, J. J. Hunt, and M. Zhou, "Diffusion policies as an expressive policy class for offline reinforcement learning," in *The Eleventh International Conference on Learning Representations*, 2023.

[12] Z. Li, N. B. Kovachki, K. Azizzadenesheli, B. Liu, K. Bhattacharya, A. Stuart, and A. Anandkumar, "Fourier neural operator for parametric partial differential equations," in *The Ninth International Conference on Learning Representations*, 2021.

[13] X. Liu, B. Xu, S. Cao, and L. Zhang, "Mitigating spectral bias for the multiscale operator learning," *Journal of Computational Physics*, vol. 506, p. 112944, 2024.

[14] H. Liu and L. Lyu, "Co-design of transmission and control in industrial IoT systems," in *2024 IEEE/CIC International Conference on Communications in China (ICCC)*. IEEE, 2024, pp. 615–620.

[15] C. Zhang, Y. Shi, and Y. Chen, "BEAR: Physics-principled building environment for control and reinforcement learning," in *Proceedings of the 14th ACM International Conference on Future Energy Systems*. ACM, 2023, pp. 66–71.

[16] *Thermal Environmental Conditions for Human Occupancy*, American Society of Heating, Refrigerating and Air-Conditioning Engineers, Atlanta, GA, 2023, ANSI/ASHRAE Standard 55-2023.

[17] J. Ho, A. N. Jain, and P. Abbeel, "Denoising diffusion probabilistic models," in *Advances in Neural Information Processing Systems*, vol. 33. Curran Associates, Inc., 2020, pp. 6840–6851.

[18] S. Fujimoto and S. S. Gu, "A minimalist approach to offline reinforcement learning," in *Advances in Neural Information Processing Systems*, vol. 34. Curran Associates, Inc., 2021, pp. 20 132–20 145.

[19] J. R. Vazquez-Canteli, G. Henze, and Z. Nagy, "MARLISA: Multi-agent reinforcement learning with iterative sequential action selection for load shaping of grid-interactive connected buildings," in *Proceedings of the 7th ACM International Conference on Systems for Energy-Efficient Buildings, Cities, and Transportation*. ACM, 2020, pp. 170–179.

[20] L. Yu, Y. Sun, Z. Xu, C. Shen, D. Yue, T. Jiang, and X. Guan, "Multi-agent deep reinforcement learning for HVAC control in commercial buildings," *IEEE Transactions on Smart Grid*, vol. 12, no. 1, pp. 407–419, 2021.

[21] W. Xue, N. Jia, and M. Zhao, "Multi-agent deep reinforcement learning based HVAC control for multi-zone buildings considering zone-energy-allocation optimization," *Energy and Buildings*, vol. 329, p. 115241, 2025.